

MINGYUE GUO

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EDUCATION

Texas A&M University, Texas, U.S.

January 2023 - present

Ph.D. candidate in Mechanical Engineering

GPA 4.0/4.0

- **Dissertation:** Developing practical learning-based controls for building energy systems
- **Advisor:** Dr. Zheng O'Neill, PhD, PE, FASHRAE, FIBPSA

Tongji University, Shanghai, China

September 2019 - March 2022

M.S. in HVAC & gas engineering

overall GPA 4.37/5

- **Thesis:** A hybrid energy consumption prediction model for office buildings based on heterogeneous data (**Excellent Master's Thesis Award**)
- **Advisor:** Peng Xu, PhD, PE, MBA, FIBPSA

Chongqing University, Chongqing, China

September 2015 - June 2019

B.E. in Built Environment; minor in Business Administration

GPA 3.69/4; rank:2/113

- **Thesis:** Preliminary Study on the automatic design of HVAC system based on BIM
- **Supervisor:** Nan Li, Peng Xu

PROFESSIONAL EXPERIENCE

Research Assistant Texas A&M University

2023 - present

mentor(s): Zheng O'Neill

Research included: occupancy-centric control field testing and results analysis, large-scale building energy simulation and data analysis, and intelligent controllers (e.g., MPC, RL, and LLM) for efficient energy management.

Machine Learning Research Intern Hewlett Packard Enterprise

May - Aug. 2025

mentor(s): Soumyendu Sarkar, Avisek Naug

Developed reinforcement learning control for data center liquid cooling systems.

Graduate Research Intern Oak Ridge National Laboratory

Jun. - Aug. 2024

mentor(s): Sen Huang, Jamie (Jianming) Lian

Developed control-oriented models and conducted field testing for model-predictive control strategies to optimize energy costs in a grid-interactive commercial building while maintaining thermal comfort.

Research Assistant Hong Kong University of Science and Technology

Sep. - Dec. 2022

Conducted research on implementing model predictive control to optimize the heating, ventilation, and air conditioning (HVAC) system of a commercial building.

Research Assistant Tongji University

Sep. 2019 - Mar. 2022

mentor(s): Peng Xu

Developed a hybrid energy prediction model using data driven method.

Conducted research on automatic BIM to BEM and HVAC system sizing.

Assistant Designer CMCU Engineering Co. Ltd. Chongqing, China

Jul. - Aug. 2017

Assisted in the design of the HVAC system for a high-rise residential building in Guizhou, China.

JOURNAL PUBLICATIONS

Equal Contribution.

- [1] **M. Guo**, Z. O'Neill, Z. Yang, J. Wen, W. Sun, Cleanroom environmental control systems: review of research landscape and control strategies, *Energy and Buildings* 365 (2026) 117639. <https://doi.org/10.1016/j.enbuild.2026.117639>.
- [2] Z. Pang#, **M. Guo**#, Z. O'Neill, B. Smith-Cortez, Z. Yang, B. Dong, A longitudinal field study of sensor-driven occupancy-centric HVAC controls in an office building, *Energy and Buildings* 351 (2026) 116693. <https://doi.org/10.1016/j.enbuild.2025.116693>.
- [3] **M. Guo**, Y. Choi, S.-M. Cheong, Z. O'Neill, Current and Future Residential Electricity Demand Using Large-Scale Smart Meter Data in a Changing Climate, *Sustainable Cities and Society* (2025) 106623. <https://doi.org/10.1016/j.scs.2025.106623>.
- [4] **M. Guo**, B. Liu, S. Huang, J. Dong, J. Lian, Z. O'Neill, Deploying advanced supervisory control strategies to small- and medium-sized commercial buildings: Case study and lessons learned, *Energy and Buildings* (2025) 115710. <https://doi.org/10.1016/j.enbuild.2025.115710>.
- [5] Z. Pang, **M. Guo**, Z. O'Neill, B. Smith-Cortez, Z. Yang, M. Liu, B. Dong, Long-Term field testing of the accuracy and HVAC energy savings potential of occupancy presence sensors in A Single-Family home, *Energy and Buildings* 328 (2025) 115161. <https://doi.org/10.1016/j.enbuild.2024.115161>.
- [6] **M. Guo**, B. Smith-Cortez, Z. O'Neill, T. Firsich, Z. Yang, Exploring and field-demonstrating geofence-based occupancy-centric control in residential buildings, *Energy and Buildings* 320 (2024) 114595. <https://doi.org/10.1016/j.enbuild.2024.114595>.
- [7] M. Liu, **M. Guo**, Y. Fu, Z. O'Neill, Y. Gao, Expert-guided imitation learning for energy management: Evaluating GAIL's performance in building control applications, *Appl. Energy* 372 (2024) 123753. <https://doi.org/10.1016/j.apenergy.2024.123753>.
- [8] Z. Pang, **M. Guo**, B. Smith-Cortez, Z. O'Neill, Z. Yang, M. Liu, B. Dong, Quantification of HVAC energy savings through occupancy presence sensors in an apartment setting: Field testing and inverse modeling approach, *Energy and Buildings*. 302 (2024) 113752. <https://doi.org/10.1016/j.enbuild.2023.113752>.
- [9] D. Wang, M. Li, **M. Guo**, Q. Shi, C. Zheng, D. Li, S. Li, Z. Wang, Modelling variable refrigerant flow system for control purpose, *Energy and Buildings*. 292 (2023) 113163. <https://doi.org/10.1016/j.enbuild.2023.113163>.
- [10] H. Zhong, **M. Guo**, Y. Wang, Z. Wang, Quantify the magnitude and energy impact of overcooling in a sub-tropical campus building, *Build. Environ.* (2023) 110033. <https://doi.org/10.1016/j.buildenv.2023.110033>.
- [11] Y. Chen, **M. Guo**, Z. Chen, Z. Chen, Y. Ji, Physical energy and data-driven models in building energy prediction: A review, *Energy Rep.* 8 (2022) 2656–2671. <https://doi.org/10.1016/j.egyrs.2022.01.162>. [**Highly Cited Paper**]
- [12] **M. Guo**, P. Xu, T. Xiao, R. He, M. Dai, S.L. Miller, Review and comparison of HVAC operation guidelines in different countries during the COVID-19 pandemic, *Build. Environ.* 187 (2021) 107368. <https://doi.org/10.1016/j.buildenv.2020.107368>. [**Highly Cited Paper**]

MANUSCRIPTS IN PREPARATION

- [1] **M. Guo**, Z. Qi, S. Zhan, Q. Zhu, Z. O'Neill, Benchmarking Intelligent Controls in Building HVAC Systems: an Extensible and Modular Framework and a Case Study

- [2] **M. Guo**, Z. O'Neill, Z. Yang, A Co-simulation Platform for Energy-Saving Evaluation of HVAC Control Strategies in Cleanrooms
- [3] **M. Guo**, Y. Choi, Z. O'Neill, S., Cheong, What Matters Most? Feature Importance of Demographics, Tree Canopy, and Temperature for ZIP Code-Level Energy Use Intensity

CONFERENCE PAPERS

- [1] H.H. Ghoneim, **M. Guo**, Z. Yang, Z. O'Neill, Assessing Room-Level Thermal Variation in a Single-Family Home Through a Long-Term Thermostat Reference and Temperature Distribution Analysis, 2026 ASHRAE Winter Conference, Las Vegas, United States, Jan. 31-Feb. 4, 2026. <https://doi.org/10.63044/w26gho24>.
- [2] L. Pinheiro, **M. Guo**, Z. Wang, Z. Pang, Z. O'Neill, X. Lu, Uncertainty Quantification of Occupancy Sensor Errors on Energy Savings and Thermal Comfort in Residential Homes, 2026 ASHRAE Winter Conference, Las Vegas, United States, Jan. 31-Feb. 4, 2026.
- [3] **M. Guo**, Z. Pang, Z. O'Neill, B. Smith-Cortez, Z. Yang, B. Dong, Long-Term Field Testing of Occupant Counting Sensors and Occupancy-Centric HVAC Controls in an Office Building, 2025 ASHRAE Winter Conference, Orlando, United States, February 8-12, 2025.
- [4] L. Pinheiro, **M. Guo**, Z. Pang, Z. O'Neill, Quantifying Energy Savings of Occupancy-Centric HVAC Controls Utilizing Longitudinal Occupancy Sensing Data and Calibrated Energy Simulations, 2025 ASHRAE Winter Conference, Orlando, United States, February 8-12, 2025.
- [5] **M. Guo**, Y. Fu, M. Liu, Z. O'Neill, Investigations on the Influence of Model Accuracy in Deep Reinforcement Learning Control for HVAC Application, 2024 ASHRAE Winter Conference, Chicago, United States, January 20-24, 2024.
- [6] **M. Guo**, P. Xu, H. Wang, J. Gu, Z. Chen, Building energy modelling based on building information modelling: the remaining problems and a more robust method, In Proceedings of the 18th International Building Simulation Conference of IBPSA, Shanghai, China, September 4-6, 2023: pp. 1022–1030. <https://doi.org/10.26868/25222708.2023.1412>

POSTERS AND INVITED PRESENTATION

- [1] **M. Guo**, S.-M. Cheong, Z. O'Neill, Engineering and Policy Insight into Residential Electricity Demand Resilience in a Changing Climate: Evidence from Large-Scale Smart Meter Data, oral presentation in Texas A&M Conference on Energy, College Station, United States, September 25, 2025.
- [2] **M. Guo**, Z. O'Neill, Z. Yang, C. Chang, An Integrated Approach to Modeling, Decision-making, and Control for Energy Efficient Manufacturing, poster presentation in 2025 Texas HSRUs and HSIs: STEAM Research showcase, College Station, United States, September 11, 2025.
- [3] **M. Guo**, Z. O'Neill, S. Cheong, Characterizing Residential Electricity Consumption in Harris County: Key Patterns and Demographic Influences, poster presentation in 2024 Texas Energy Summit, Austin, United States, November 19-21, 2024.
- [4] **M. Guo**, Z. Pang, Z. O'Neill, Prediction of Occupant Number and Identification of Occupant Behaviors via Learning Algorithms – A Case Study in a Small Office Building. Seminar in Proceedings of the 2023 ASHRAE Annual Conference, Tampa, U.S., June 24-28, 2023.

SELECTED RESEARCH PROJECTS

1. **Texas grid impact of replacing existing residential AC with heat pumps** Jun. - Nov. 2026
Funded by Daikin, lead graduate researcher at TAMU
 - Simulate the Texas grid impacts of replacing existing air-conditioning systems with federally compliant minimum-efficiency heat pumps and inverter-driven heat pumps using ResStock and EnergyPlus.

- Calibrate the EnergyPlus models of single-stage and inverter-based heat pumps using field test data.

2. An Integrated Approach to Modeling, Decision-Making and Control for Energy Efficient Manufacturing Sep. 2023 - present

Funded by National Science Foundation (NSF), lead graduate researcher in TAMU

- Developed and validated control-oriented models for the HVAC systems of semiconductor cleanrooms.
- Design and implement a decision-making and adaptive control for AHUs

3. Large-Scale Smart Meter Data Analysis May 2024 - present

Funded by Texas A&M University

- Data cleaning for real-world smart meter data from over 2 million consumers in the Greater Houston Area, Texas.
- Analyzed the energy resilience in a changing climate till 2090 by developing data-driven models.
- Correlated electricity consumption with demographic factors and tree canopy.

4. Quantification of HVAC Energy Savings for Occupancy Sensing in Buildings Through an Innovative Testing Methodology Jan. 2023 - Aug. 2024

Funded by Advanced Research Projects Agency-Energy (ARPA-E), research assistant in TAMU

- Maintained, and troubleshoot field-testing systems in a single-family home and a commercial building.
- Analyzed the sensor accuracy, energy savings, and thermal comfort in three field tests.
- Extracted typical occupancy schedule through field-measured data using the K-means clustering algorithm.

5. Development and Validation of Smart Building Technology Modules for Academic and Professional Education Jan. 2023 - Dec. 2025

Funded by U.S. Department of Energy (DOE), research assistant in TAMU

- Polished and proofread the educational modules

6. Target-controlled Feedforward Management Technology for Green Buildings 2019 - 2021

Funded by the "13th Five-Year Plan" Key Research Program, China; research assistant in Tongji Univ.

- Developed an automatic tool to convert BIM (Building Information Modeling) in Revit to BEM (Building Energy Modeling) in EnergyPlus through an intermediate gbXML file.
- Enabled automatic gbXML check and IDF enrichment to ensure the success of the BIM to BEM conversion.

7. Automatic HVAC design 2020 - 2021

Funded by Tongji University Architectural Design Institute Co., Ltd.; research assistant in Tongji Univ.

- Developed an automatic configuration tool for the AHU, FCU, and chiller sizing in Python

TEACHING EXPERIENCE

Teaching Assistant @Tongji Univ. 2020 - 2021

Business Administration in the Master of Engineering Management (MEM) program

Managed attendance records, coordinated group discussion schedules, and monitored the progress of course projects.

Research Mentor @TAMU

Mentored undergraduate and early-stage Ph.D. students in research projects, including research idea development, research tool utilization, and academic writing, resulting in co-authored conference publications (e.g., C[1]).

HONORS AND AWARDS

2025	J. Mike Walker '66 Impact Award in Mechanical Engineering, TAMU	
2025	Melbern G. Glasscock '59 Endowed Graduate Fellowship in Mechanical Engineering, TAMU	
2025	ASHRAE Graduate Grant-in-Aid Award	
2024	Brenda and Jerry Gray '62 Fellowship in Mechanical Engineering, TAMU	
2024-2025	Graduate Travel Award from Department of Mechanical Engineering	
2022	Outstanding Graduate of Shanghai Municipality	1%
2021	Outstanding Students, Tongji University	2%
2021	National Scholarship, Ministry of Education of the People's Republic of China	0.2%
2020	3 rd prize of Yada Scholarship, Tongji University	
2019	Outstanding Graduate of Chongqing Municipality	1%
2017-2019	National Encouragement Scholarship for 3 times, Chongqing University	5%
2018	Outstanding Students, Chongqing University	5%
2017	Excellent League Member, Chongqing University	5%

COMPETITION AWARDS

Global AI competition March 2022
Organizer: the Electrical and Mechanical Services Department (EMSD) of the Government of Hong Kong Special Administrative Region (HKSAR) and the Guangdong Provincial Association for Science and Technology

- AWARD: **Best use of Tencent cloud** award, **Huawei most innovative use of data** award and **Gold award** (Team leader, top 1 in the leaderboard)
- Description: Predict hourly cooling load of the next three months with historical data in about previous a year and a half.

IFLYTEK A.I. Developer Competition: Demand Strategy Optimization Based on Flexible Load June to October 2022
Organizer: Sun Power

- AWARD: The First Place (First author, First prize)
- Description: Rearrange the flexible load and battery schedule for 5 months in 5-minute intervals to reduce the electricity cost.

PROFESSIONAL SOCIETIES AND SERVICE

- Reviewer of *Science and Technology for the Built Environment* journal and ASHRAE conference paper, 2023 to present.
- Student Member of ASHRAE and ASHRAE Texas A&M Student Chapter member, 2023 to present.
- Officer of the American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE) TAMU Student Branch, Fall 2023 - Spring 2025.
- Volunteer for the revision of *ASHRAE Handbook—HVAC Applications* chapter 43 Supervisory Control Strategies and Optimization

REFERENCES

Available upon request

Last Updated: Jul., 2026